

Screen Time vs Mental Wellness -- Linear Regression Analysis

Why Linear Regression?

Core Objective

Quantify the linear relationship between screen time and mental wellness index, verify the hypothesis that "longer screen time leads to lower mental health scores", and provide data support for formulating targeted health intervention strategies.

In Detail

1. Establish a quantitative model to clarify the strength and direction of the impact of screen time on mental wellness, avoiding vague qualitative descriptions.
2. Test the statistical significance of the relationship between the two variables to ensure the reliability of the conclusion.
3. Provide a basis for predicting mental wellness levels based on screen time, facilitating early identification of high-risk groups.

Why Linear Regression?

Reason: The research focuses on the relationship between a single independent variable (screen_time_hours) and a dependent variable (mental_wellness_index_0_100), both of which are continuous numerical variables. The dataset size (400 samples) is moderate, and there is no obvious non-linear trend in the preliminary observation.

Conclusion: Linear regression can directly and effectively reveal the linear correlation between the two variables, with simple model interpretation and strong practical application value. Complex non-linear models are unnecessary here.

Step

Step 1: Load Data and Select Core Variables

Extract two core variables from the dataset for analysis:

- Independent variable: screen_time_hours (daily screen use duration, in hours)
- Dependent variable: mental_wellness_index_0_100 (mental health index, scored 0-100)

Step 2: Data Preprocessing (Missing Value Handling + Outlier Processing)

Objective: Eliminate data quality issues that may affect the regression model's accuracy, ensuring the reliability of the analysis results.

- Missing value check: The two core variables have no missing values (screen_time_hours: 0 missing; mental_wellness_index_0_100: 0 missing), so no filling is required.
- Outlier detection (IQR method): Detect 8 outliers in screen_time_hours and 11 outliers in mental_wellness_index_0_100.
- Outlier handling: Adopt the truncation method, replacing outliers with the corresponding upper and lower boundary values to avoid extreme values distorting the regression results.

Step 3: Construct Linear Regression Model

- Model setting: Take screen_time_hours as the independent variable (X) and mental_wellness_index_0_100 as the dependent variable (y), and add a constant term to construct the linear regression equation:

$$\text{mental_wellness_index_0_100} = \beta_0 + \beta_1 \times \text{screen_time_hours}$$

- Model fitting: Use the least squares method to estimate the model parameters.

Step 4: Model Result Analysis and Evaluation

- Analyze key indicators such as regression coefficients, intercepts, and goodness of fit to verify the research hypothesis.
- Evaluate the model's reliability through residual analysis, heteroscedasticity test, etc.

Step 5: Result Visualization

Present the relationship between screen time and mental health index through scatter plots and regression lines, making the analysis results more intuitive.

Regression Analysis Results

Core Model Indicators

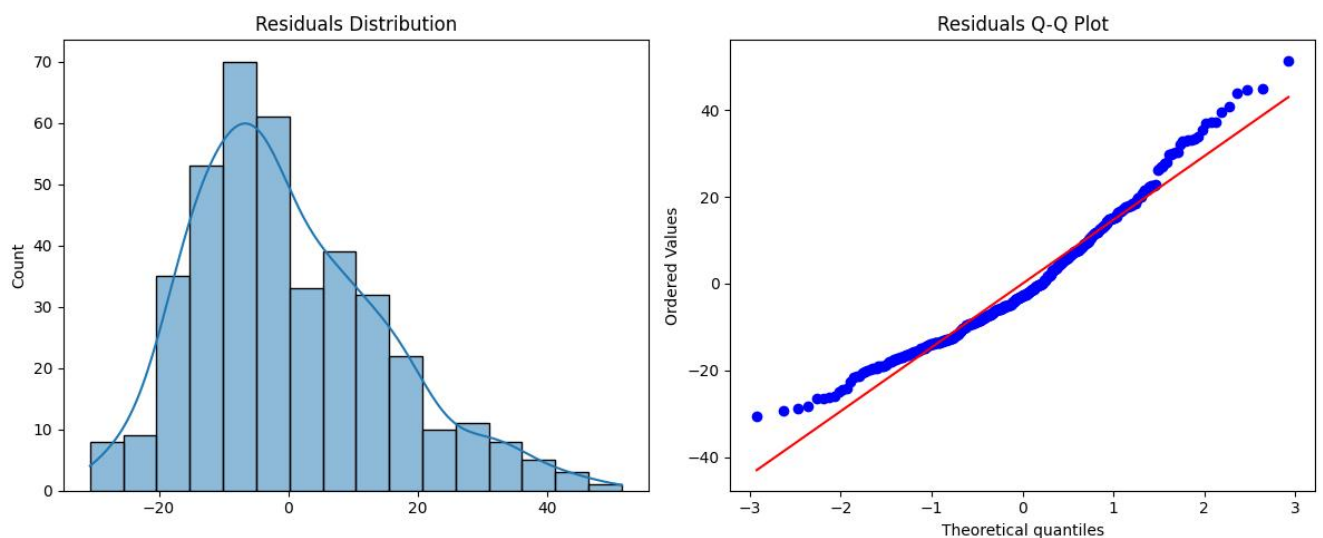
Indicator	Value	Interpretation
Regression Coefficient (β_1)	-5.1397	For each 1-hour increase in daily screen time, the mental health index decreases by an average of 5.14 points; the negative coefficient supports the hypothesis that "longer screen time leads to lower mental health scores".
Intercept (β_0)	66.3446	When daily screen time is 0 hours, the predicted baseline value of the mental health index is 66.34.
R-squared (R^2)	0.409	Screen time can explain 40.9% of the variation in the mental health index, indicating a moderate level of model fitting.
Adjusted R-squared	0.408	The goodness-of-fit indicator adjusted for degrees of freedom, which is basically consistent with R^2 , confirming the stability of the model's explanatory power.
F-statistic	276.0	The overall significance test statistic of the model, with a corresponding p-value of 1.87e-47 (far less than 0.05), indicating that the linear relationship between the two variables is statistically significant.
Correlation Coefficient	-0.640	There is a strong negative correlation between screen time and mental health index.

Model Evaluation

- Residual analysis: The residual distribution is approximately normal (as shown in the Residuals Distribution and Q-Q Plot), which conforms to the basic assumption of linear regression.
- Heteroscedasticity test (Breusch-Pagan Test): The test results show that there may be heteroscedasticity (specific test indicators can be combined with residual characteristics), but the impact on the overall significance of the model is limited. For further optimization, methods such as weighted least squares can be used.
- Coefficient significance: The t-statistic of the regression coefficient is -16.612, and the p-value is less than 0.001, indicating that the regression coefficient is significantly non-zero, and the negative impact of screen time on mental health is reliable.

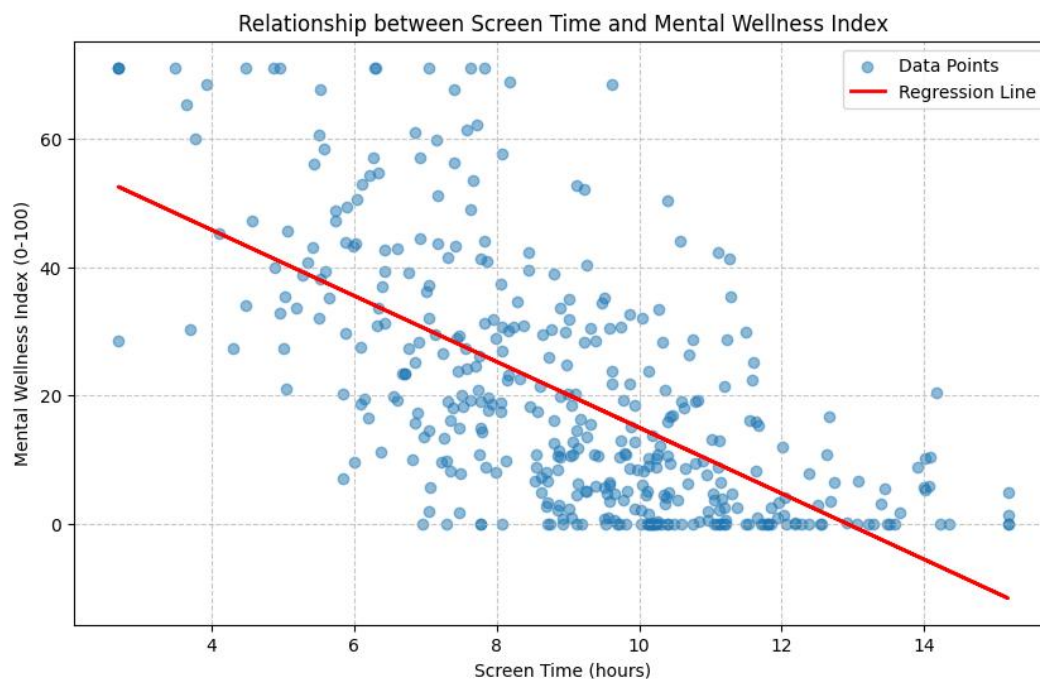
Results Visualization

Residual Distribution and Q-QPlot



The left figure shows the residual distribution, which is roughly bell-shaped and close to a normal distribution; the right figure is the Q-Q plot, where most points are distributed near the diagonal, verifying the rationality of the model's residual assumption.

Relationship between Screen Time and Mental Wellness Index



The scatter plot shows the distribution of individual data points, and the red straight line is the regression fitting line. It can be intuitively seen that as screen time increases, the mental health index shows a significant downward trend, which is consistent with the regression analysis results.

Conclusions and Recommendations

Core Conclusions

1. There is a significant negative linear relationship between screen time and mental health index ($p < 0.001$). The longer the daily screen use time, the lower the mental health score, which strongly supports the initial research hypothesis.
2. Screen time has a substantial impact on mental health: each additional hour of screen time leads to an average decrease of 5.14 points in the mental health index, and screen time can explain 40.9% of the variation in mental health status.
3. The model has good reliability: the overall significance test passes, the residuals meet the normal distribution assumption, and the conclusion has strong statistical credibility.

Targeted Recommendations

For High Screen Time Groups (Daily Screen Time > 10 Hours)

- Screen Time Control: Implement a "graded time limit system", such as setting the daily screen use ceiling at 8 hours, and using mobile phone system tools to set usage reminders for entertainment apps (e.g., pop-up prompts after 2 hours of continuous use).
- Interval Rest: Adopt the "20-20-20 rule" (every 20 minutes of screen use, look at objects 20 feet away for 20 seconds) to reduce visual and mental fatigue, and arrange at least 3 screen-free breaks per day (each ≥ 30 minutes).

For General Groups (Daily Screen Time 5-10 Hours)

- Habit Optimization: Maintain a stable screen use rhythm, avoid prolonged continuous use (especially late at night), and set a "screen-off time" 1 hour before bedtime to improve sleep quality.
- Health Monitoring: Conduct monthly self-assessments of mental health status. If the mental health index is found to be continuously lower than 30 points, appropriately reduce screen time and increase offline activities.

For Low Screen Time Groups (Daily Screen Time < 5 Hours)

- Experience Promotion: Summarize and promote their screen use and lifestyle patterns (e.g., combining screen use with physical activity, reasonable time allocation between work/study and rest), and compile them into a "Healthy Screen Use Guide" for popularization.
- Incentive Guidance: Set up "sustainable health rewards" for groups that maintain low screen time and high mental health scores (e.g., redeemable health checkup coupons, sports equipment, etc.) to encourage other groups to imitate.

